MORTGAGE QA/QC ENGINE — COMPILER PIPELINE

The complete picture: six specs, one pipeline, three gating layers

Where every piece of this session's work fits — what's implemented, what's specced but not built, and the one sequencing dependency the judge review surfaced: real-impact measurement is blocked until two specific things ship, in order.

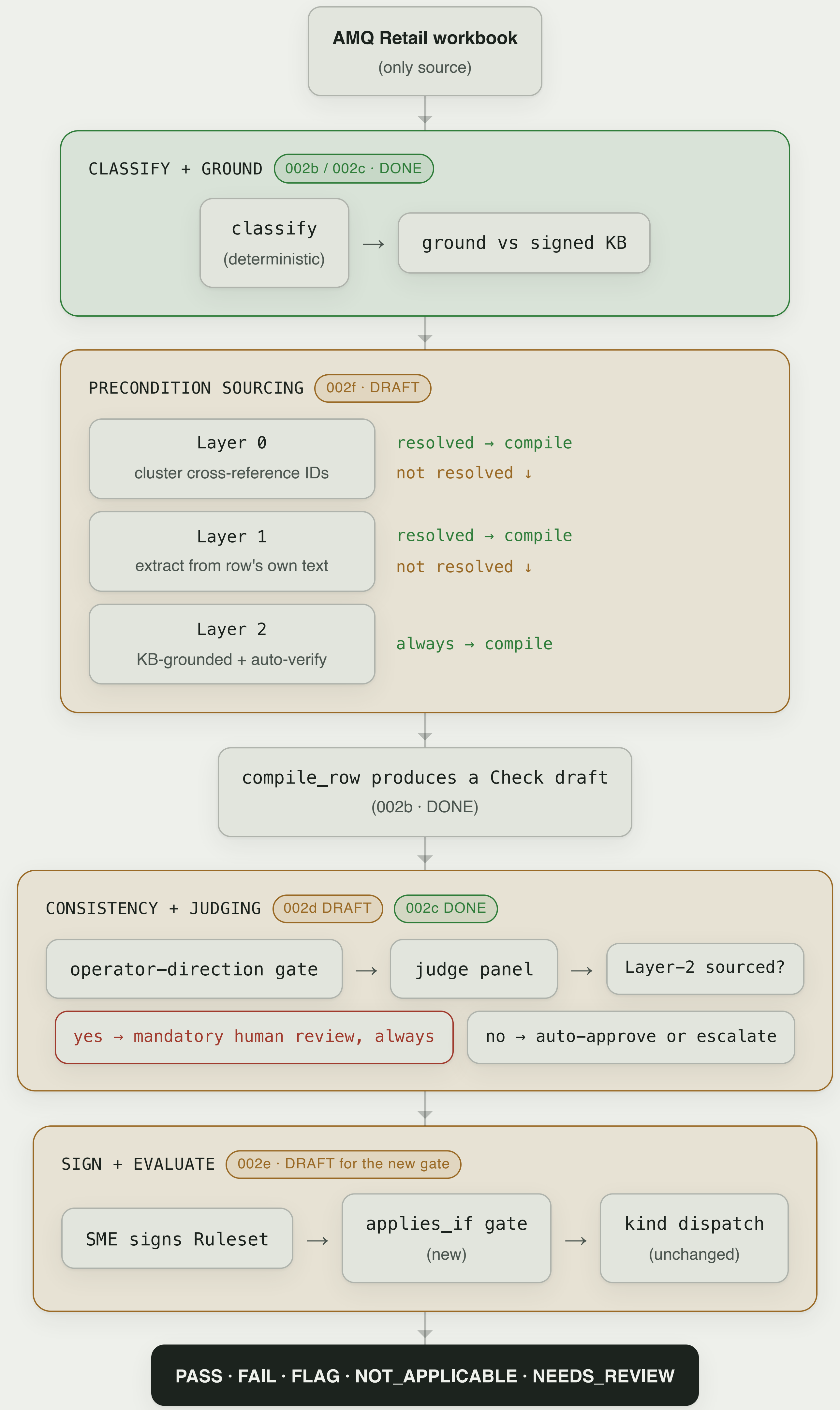
Implemented, tested · Specced, not built · Decided, not yet executed

The six specs, at a glance

SPEC	WHAT IT OWNS	STATUS
`002a`	Compile-fidelity spike — proved the LLM-authoring step works at all	Implemented
`002b`	The core compiler pipeline — `compile_llm.py`, one Bedrock call per row	Implemented
`002c`	Signed KB + grounding retrieval + multi-model judge panel	Implemented (164 tests)
`002d`	Operator-direction consistency gate — fixes the LTV/MI backwards-boundary bug	Draft
`002e`	`Check.applies_if` schema + engine gate — mortgage-specific consumer	Draft
`002f`	Precondition ontology layer (0/1/2) — the reusable extraction mechanism	Draft

End-to-end pipeline, with the new pieces layered in

Same compile-then-run spine as always — three new stages inserted where the SME call and the judge review found real gaps.



The sequencing gate the judge review surfaced

Layer 0's clustering result (24 IDs, 3,255 rows) is real and independently verified — it came straight from the raw workbook. What's NOT yet valid is measuring its real-world defect impact, because the only compiled ruleset that exists today conflates three unrelated problems.

```
002d ships → Retail–only recompile → clean ruleset exists →
real–impact measurement becomes valid
```

Why the current ruleset can't answer that question yet: it mixes the now-excluded Private Bank workbook, and it carries the known, unfixed operator-inversion bug (45+ suspects). Testing "how many of the 3,255 rows currently misfire" against it today would tangle three unrelated bugs into one number — confounded evidence, not clean evidence, in either direction.

What's already true, independent of that sequencing: the decoding mechanism itself (clustering a workbook's own cross-reference column instead of assuming it's opaque) is proven against real data, reproducible, and zero-cost — that finding doesn't wait on anything. Only the "how much does this actually matter on real loans" number waits on the two steps above.